

Question 1: What is a conditional statement in programming?

A conditional statement is a feature in programming that allows the code to make decisions and execute different blocks of code based on whether a condition is true or false.

Question 2: What does an if statement do?

An if statement checks a condition. If that condition is true, it executes the block of code inside it. If the condition is false, it skips that code.

Question 3: What is the purpose of an else block?

The purpose of an else block is to provide an alternative path. It runs a backup block of code when the main if condition turns out to be false.

Question 4: When is an else block executed?

An else block is executed only when all the preceding if (and else if) conditions evaluate to false.

Question 5: What is a Boolean expression?

A Boolean expression is any statement or condition that evaluates to either exactly true or false.

Question 6: Write the syntax of a simple if statement.

```
C++  
if (condition) {  
    // code to execute if condition is true  
}
```

Question 7: What operator is commonly used to test equality?

The double equal sign (==) operator is used to check equality.

Question 8: What does the modulus (%) operator return?

The modulus operator returns the remainder left over after dividing one number by another.

Question 9: How can you check if a number is even?

A number is even if it leaves a remainder of 0 when divided by 2.

Condition: `num % 2 == 0`

Question 10: How can you check if a number is odd?

A number is odd if it leaves a remainder of 1 (or not 0) when divided by 2.

Condition: `num % 2 != 0`

Question 12: What is a nested if statement?

A nested if statement is an if statement placed entirely inside another if or else block.

Question 13: Can an if statement exist without an else?

Yes, an if statement can easily exist on its own without requiring an else block.

Question 14: What is the output when a false condition is tested in an if statement without else?

No output comes from that conditional block; the program simply skips the code inside the if and moves to the next regular lines of code.

Question 15: What is the role of braces {} in C/C++ conditionals?

Braces {} are used to group multiple statements together into a single block so that the compiler knows exactly which lines belong to that condition.

Question 16: What does > mean?

Greater than operator.

Question 17: What does < mean?

Less than operator.

Question 18: What does >= mean?

Greater than or equal to operator.

Question 19: What does <= mean?

Less than or equal to operator.

Question 20: What does != mean?

Not equal to operator.

Question 21: What is a switch statement?

A switch statement is a control structure used to select and execute one choice from multiple constant options (cases) based on the value of a single expression.

Question 22: What is a case label in a switch statement?

A case label is a specific value written inside a switch block (case value:) that the switch expression is compared against.

Question 23: Why is break used in switch?

break is used to terminate the switch block and jump out of it immediately after a matching case is successfully executed.

Question 24: What happens if break is omitted?

If a break is omitted, the execution will continuously "fall through" into the next cases and run their code as well, even if those case conditions do not match.

Question 25: What is the default case in switch?

The default case is an optional block that executes automatically when none of the defined case values match the switch expression (acting like an else).

Question 26: Can switch compare strings in C++?

No, in standard C++, a switch statement can only compare integral types (like int, char, bool) and cannot directly compare strings.

Question 27: Which statement is better for many constant choices: if-else or switch?

A switch statement is better because it looks much cleaner, is easier to read, and can be optimized better by the compiler.

Question 28: How do you test whether a value is positive?

Condition: $\text{num} > 0$

Question 29: How do you test whether a value is negative?

Condition: $\text{num} < 0$

Question 30: How do you test whether a value is zero?

Condition: `num == 0`

Question 31: Write a condition to check if age is at least 18.

Condition: `age >= 18`

Question 32: Write a condition to check if marks are above 50.

Condition: `marks > 50`

Question 33: What is decision making in programming?

Decision making is the process of deciding the execution flow of instructions based on given conditions or inputs.

Question 34: Can nested if statements have multiple levels?

Yes, you can nest if statements inside other if statements to multiple levels deep, though it makes the code harder to read.

Question 35: What is logical AND (&&)?

Logical AND combines two conditions and returns true only if both conditions are true.

Question 36: What is logical OR (||)?

Logical OR combines two conditions and returns true if at least one of the conditions is true.

Question 37: What is logical NOT (!)?

Logical NOT reverses the value of a condition (makes true into false,

and false into true).

Section 5: Complex Logic & Comparisons

Question 38: How do you check if a year is divisible by 4?

Condition: `year % 4 == 0`

Question 39: What is a leap year?

A leap year is a calendar year containing 366 days instead of 365, which happens once every 4 years to keep the calendar aligned with earth's rotation.

Question 40: What is the first condition for a leap year?

The basic condition is that the year must be perfectly divisible by 4 (`year % 4 == 0`), except for century years which have extra rules.

Question 41: How do you find the largest of two numbers?

C++

```
if (a > b) {  
    cout << "a is largest";  
} else {  
    cout << "b is largest";  
}
```

Question 42: How do you find the largest of three numbers?

C++

```
if (a >= b && a >= c) {  
    cout << "a is largest";  
} else if (b >= a && b >= c) {  
    cout << "b is largest";  
}
```

```
} else {  
    cout << "c is largest";  
}
```

Question 43: What is an if-else-if ladder?

An if-else-if ladder is a multi-way decision structure where multiple conditions are checked sequentially from top to bottom until one matches.

Question 44: What is the difference between if and switch?

if can check ranges, logical conditions (&&, ||), and dynamic expressions.

switch only checks a single variable against fixed, discrete constant values.

Question 45: Can switch use ranges directly?

No, standard C++ switch statements cannot evaluate ranges (like case 1 to 10:) directly.

Question 46: What is fall-through in switch?

Fall-through is the behavior where execution moves into the next sequential case block automatically because a break statement was not encountered.

Question 47: What data type is usually used for conditions?

The boolean (bool) data type.

Question 48: Why are conditional statements important?

They are important because without them, a program would always run the exact same steps in the exact same order, making it impossible to handle dynamic user inputs or realistic logic.

Question 49: Give one real-life example of a conditional statement.

ATM Login: If your PIN code is correct, you can withdraw money; else, the machine displays an "Incorrect PIN" message.

Question 50: Name four conditional structures covered.

Simple if

if-else block

if-else-if ladder

switch case statement